portfolio



MUHAMMAD RIFKY NAUFAL



About Me

Muhammad Rifky Naufal

I’m passionate about industrial automation, instrumentation and embedded systems with experience in PLC programming, PCB design and IoT. I won 1st place at the National Selection of WorldSkills ASEAN 2025 (Mechatronics) and 2nd place at LKS-SMK 2023 (Mechatronics). Since 2026 I built IoT-based disaster early warning prototypes at PUSDI MKPI ITS and aided practical sessions at the PLC & SCS Lab. Seeking an internship in automation, electrical, and instrumentation.

Education

INSTITUT TEKNOLOGI SEPULUH NOPEMBER

Applied Bachelor (D4) in Electrical Automation Engineering | GPA: 3.5 / 4.00
Aug 2024 - Present

Featured Research Project::

Leading hardware and embedded development for INSAMO FLOWS, an IoT-based flood early warning system - designing sensor wiring and power integration, programming ESP32-based field units, and troubleshooting connectivity issues across field deployment at PUSDI MKPI ITS.

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Hard Skills

PLC Programming
HMI Development
Electrical Wiring & Panel Control
Industrial Troubleshootin
Electrical Wiring & Panel Control
Electronics & PCB Design
Embedded Systems & IoT
Mechanical CAD Design
Circuit Design & Prototyping
Pneumatic & Electropneumatic Systems
Laser & Acrylic Cutting Design
3D Printing & Slicing
IoT Device Programming

Soft Skills

Problem Sloving
Analytical Thinking
Critical Thinking
Teamwork & Collaboration
Comunication
Continous Learning

Tools

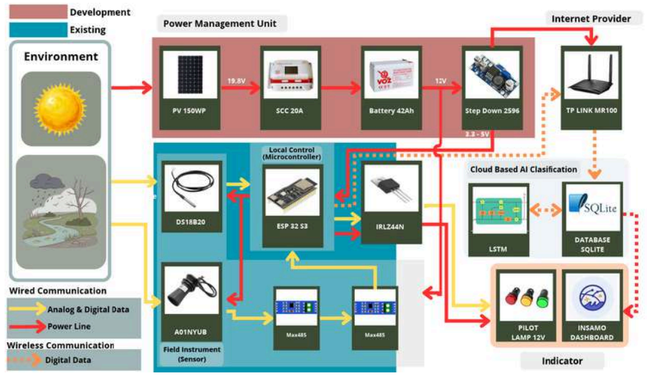


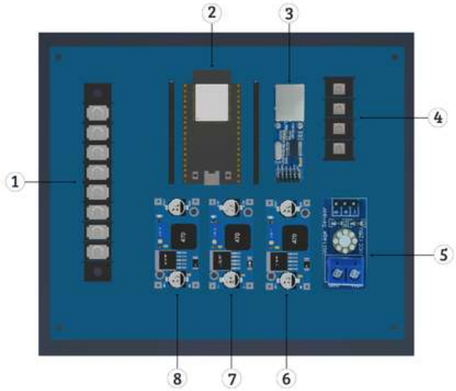
Project Experience

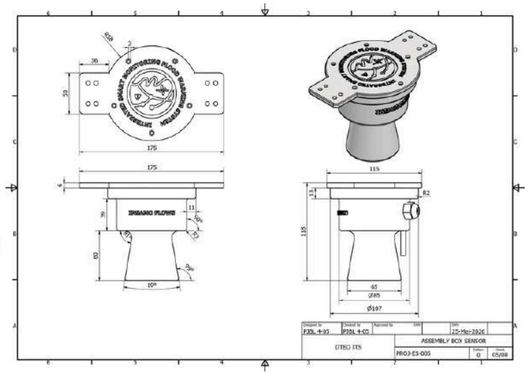
Integrated Smart Monitoring Dashboard and IoT-AI-Based Flood Warning System for Flood Mitigation, ISPF-British Council Program

ESP32-S3 | A01NYUB Ultrasonic Sensor | DS18B20 Temp Sensor | RS485 | Web Dashboard

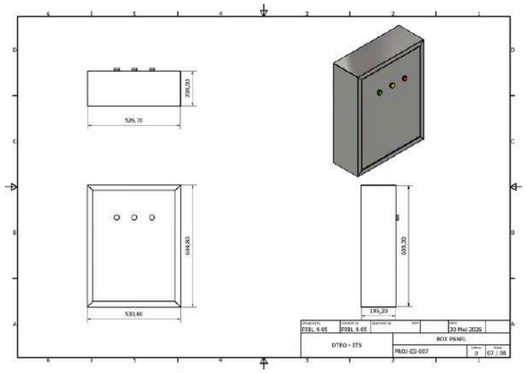
RS485 sensor integration | embedded prototyping | Comunication

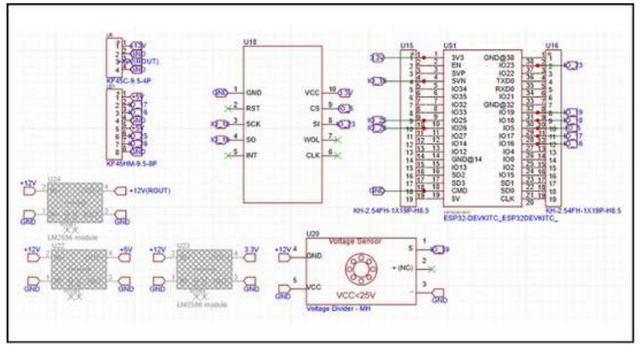


















Description

A flood monitoring system that sends water level and temperature data from field sensors to a web dashboard. An LSTM model predicts the next water level from previous time-series data. The project was developed at PUSDI MKPI ITS under the ISPF-British Council program by a team of 3.

Key Insight

- Designed the schematic and PCB routing for a flood monitoring device built on an **ESP32-S3**, with an **A01NYUB** ultrasonic sensor and a **DS18B20** temperature sensor over **RS485.**
- Created the mechanical design of the device.
- Programmed the ESP32 with a **Kalman filter** to smooth sensor data.
- Implemented **OTA (Over-the-Air)** updates so firmware can be updated without a cable.
- Assisted with wiring and troubleshooting.
- The system uses an **LSTM** model to predict water level from previous time-series data.

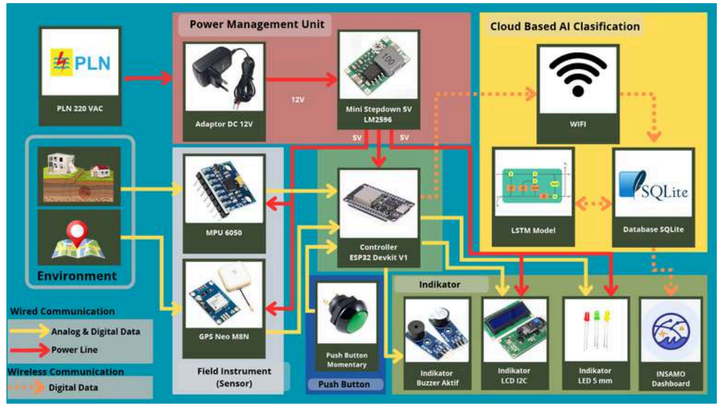
Outcome

The prototype was tested at the ITS Research Center building. The LSTM model predicted water level with a **MAPE of 2.21%**

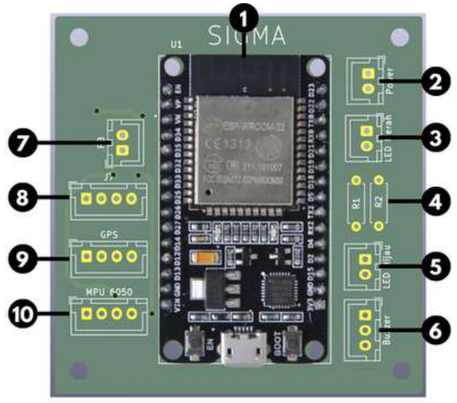
INSAMO SIGMA: IoT-Based Three-Axis Vibration Detection and Geolocation System to Support Earthquake Disaster Mitigation

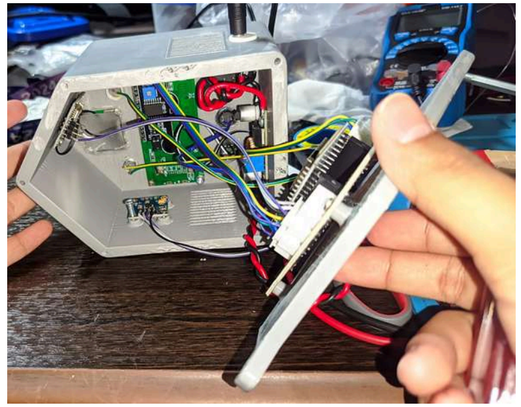
Preprocessing | Feature Engineering | Modeling | Optimization

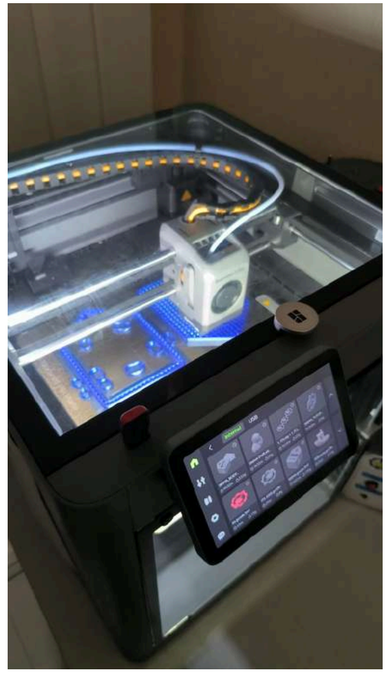
Analytical Thinking | Problem Sloving | Continous Learning

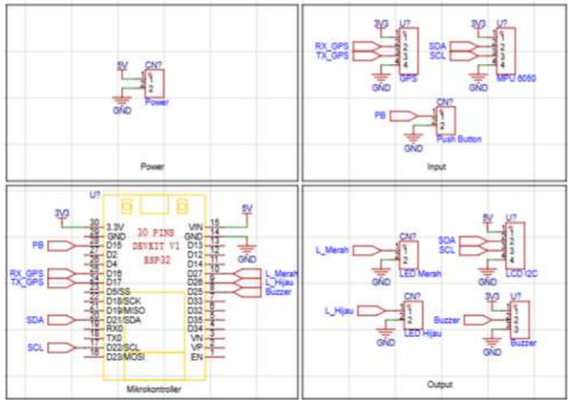




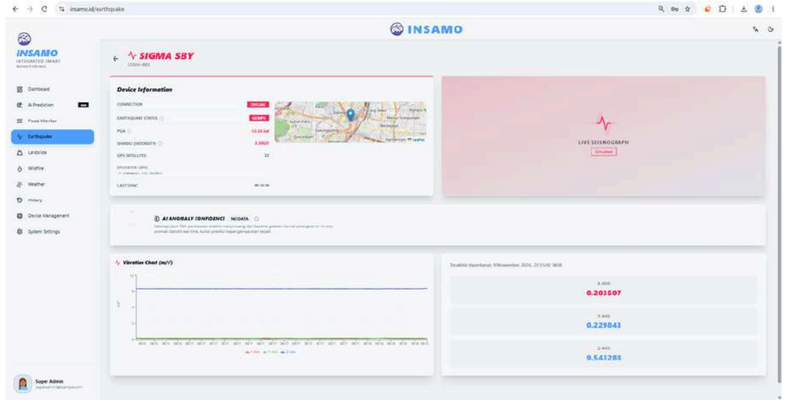












Description

A system that detects three-axis vibration and sends the data, together with the device location, to a web dashboard to support earthquake disaster mitigation. It has three modules: the vibration sensor, embedded communication, and the geolocation dashboard.

Key Insight

- Designed a three-axis vibration sensor module on an **ESP32** Dev Module with an **MPU6050** sensor.
- Built data communication between the device and the dashboard over **HTTP**, with data stored in **SQLite**.
- Located the device with GPS and displayed it on a map in the web dashboard.
- Handled the **PCB** schematic and routing, soldering, and **3D** mechanical design.
- Assisted with wiring and troubleshooting in a team of 3.

Outcome

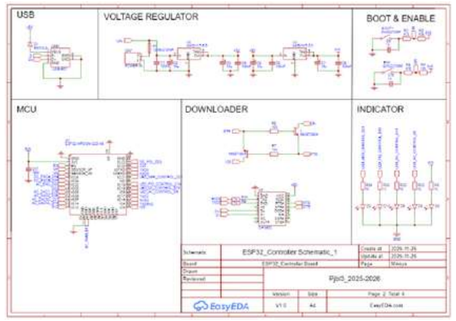
The prototype was tested for 4 months at the ITS Research Center building.

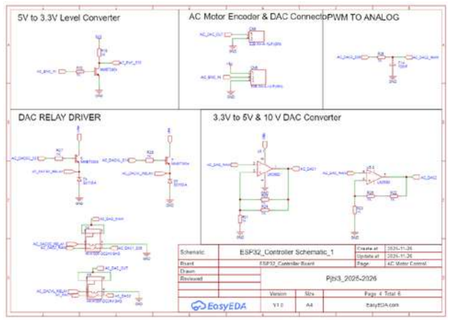
Mini Escalator with ESP32-Based PID Motor Control
Control System Designer & Embedded System Developer

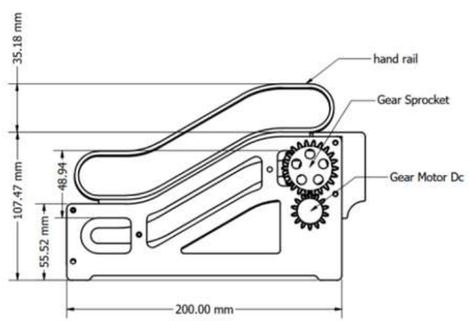
JGA12 N20 DC motor (with gearbox) | ESP32 | rotary encoder | LCD Oled | MATLAB/Simulink

Alliance Motori AC motor with VFD | Embedded C/C++ | Minertia UGTMEM-TG tachogenerator

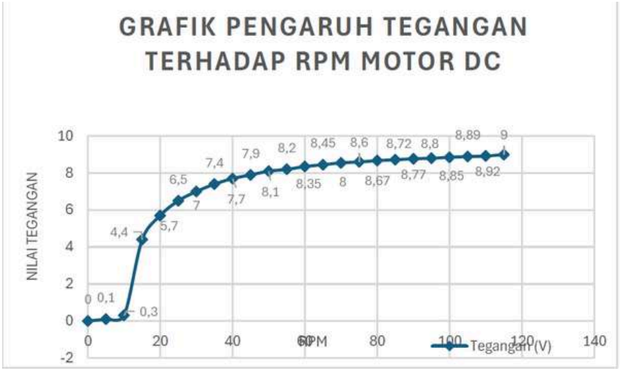
PID Control Design & Tuning | **Teamwork & Collaboration** | **Comunication** | **Embedded Programming**

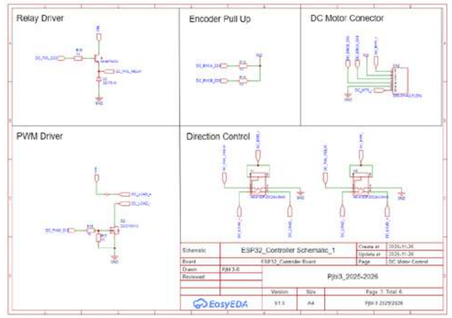


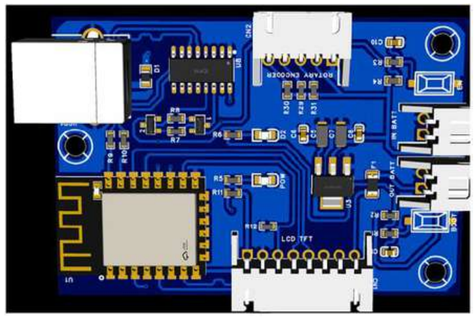


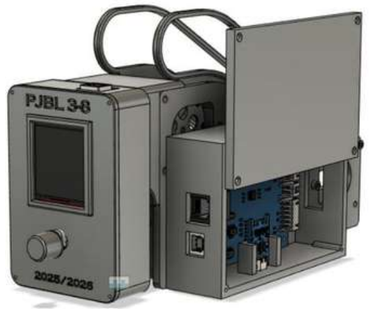




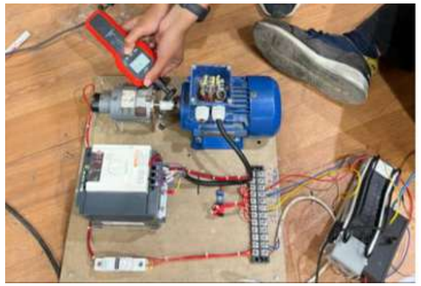












Description

A miniature escalator whose DC motor speed is controlled by a **PID algorithm** on an **ESP32**, designed to hold speed as the load changes (empty and with passengers). The RPM curve, Kp, Ki, Kd values, and system parameters appear on an LCD. The project was built as an educational application of motor control in industrial automation.

Key Insight

- Modeled the JGA12 N20 **DC motor** and an **AC motor** (1st and 2nd order) in **MATLAB/Simulink**, and simulated step response and load changes.
- Tuned **PID** parameters with Graybox Identification, Ziegler-Nichols, and Pole Cancellation.
- Programmed a discrete PID on the ESP32, using timers, counters, and hardware interrupts to read encoder pulses without blocking the main loop.
- Kept the control sampling time (Ts) constant.
- Displayed the **RPM** curve and Kp, Ki, Kd values on the LCD over SPI.
- Assembled the mini escalator **mechanics** and wiring: power module, motor driver, VFD, encoder, and display.

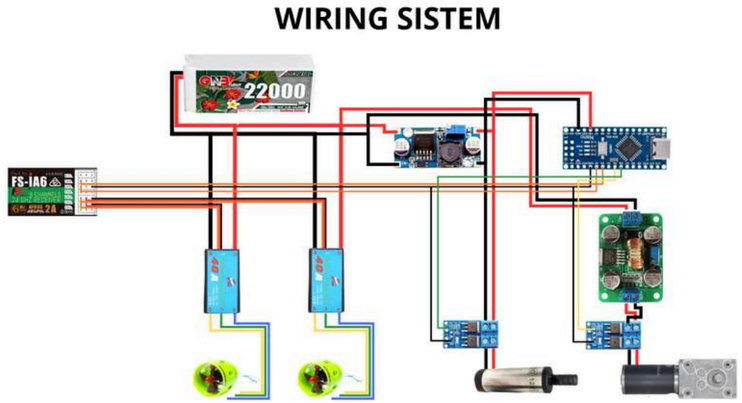
Outcome

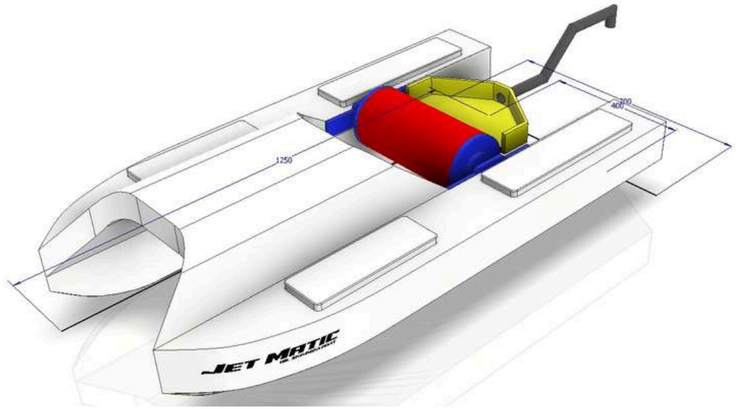
The mini escalator prototype is assembled and shows the RPM response curve on the LCD under mechanical load.

Oil Skimmer Boat - Oil Spill Collector Prototype

Arduino Nano | Motor JGA25-370 | FlySky i6 Radio Control System | Water Pump

Embeded System Programming | Electrical Wiring & Integration | Remote Control | Troubleshooting

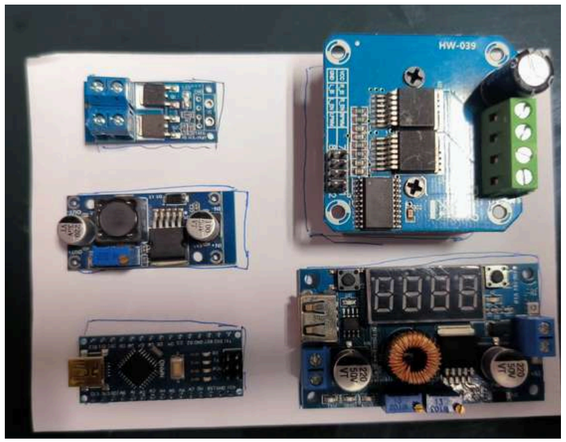


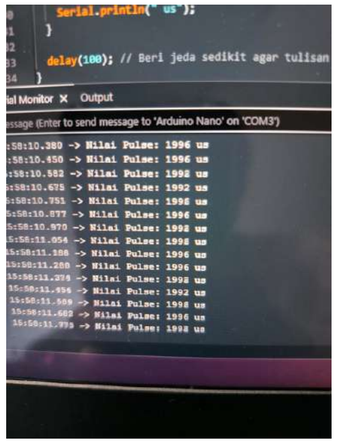
















Description

A freelance electrical and embedded systems engineering project for a small-scale prototype designed to collect spilled oil at sea. The mechanical structure was developed by a project partner, while the electrical and control systems were independently engineered.

Key Insight

- Designed and wired the complete electrical system, integrating a brushless motor for propulsion and a water pump for oil collection.
- Programmed an Arduino Nano as the main controller to handle motor control logic and radio signal communication.
- Implemented a remote control system using a FlySky i6 transmitter over radio communication, achieving a stable control range of up to 100 meters.
- · Diagnosed and resolved wiring and signal connectivity issues during prototype testing.

Outcome

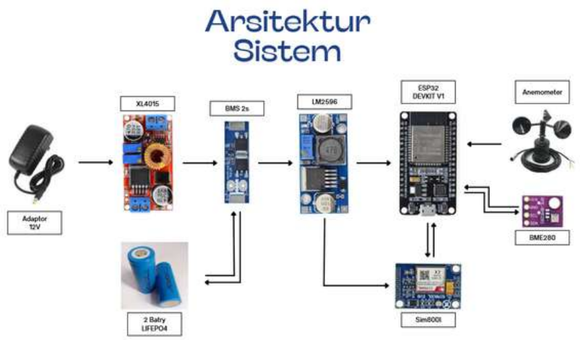
Delivered a functional prototype capable of remote-controlled navigation and oil collection, with a verified stable control range of 100 meters.

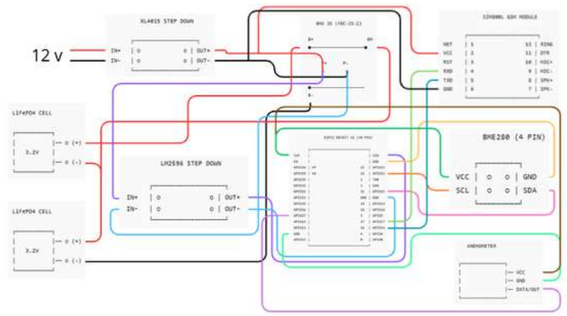
LOW-COST WEATHER MONITORING SYSTEM

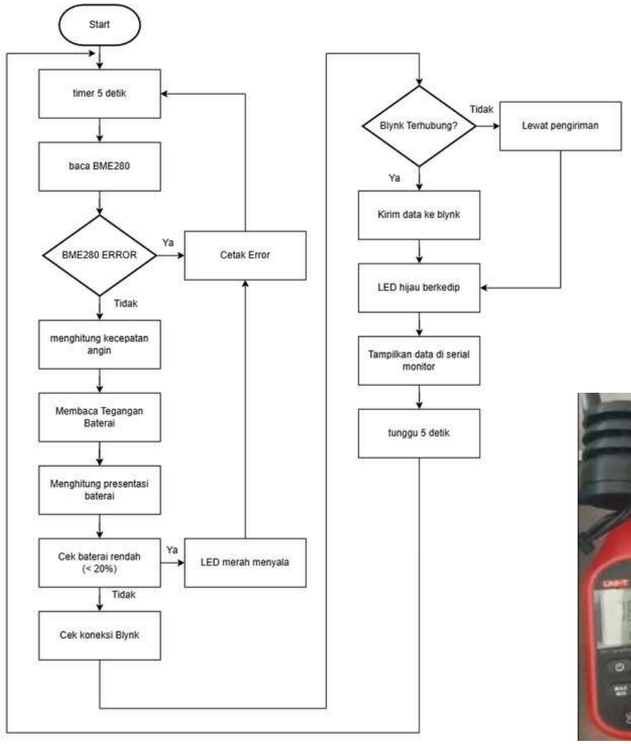
ESP32 | BME280 | Anemometer Sensor | Blynk Dashboard | Battery with BMS

Sensor Integration & Calibration | IoT Dashboard Development

Cross-Disciplinary Collaboration, Field Testing & Validation

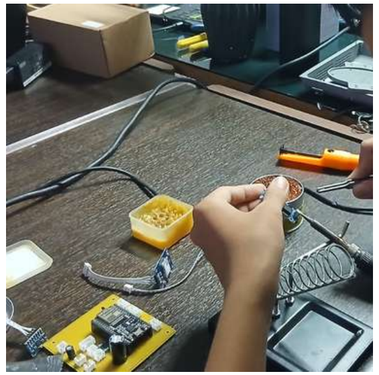


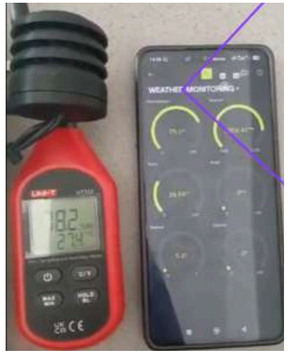




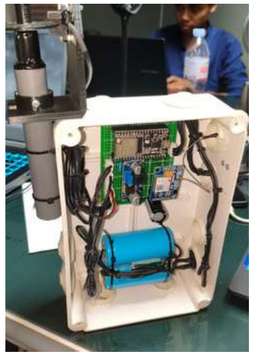












Description

A low-cost weather monitoring system built in collaboration with a lecturer from the Geomatics Engineering department at ITS, developed and tested at the Geodesy Laboratory. The system integrates environmental sensors with a real-time IoT dashboard to monitor weather conditions, with measurement accuracy validated against standard calibration instruments.

Key Insight

- Integrated ESP32 with BME280 and anemometer sensors to measure temperature and wind speed in real time.
- Developed a real-time monitoring interface using the Blynk dashboard, enabling remote weather data visualization.
- Designed the power system using a battery with BMS 1S for stable, portable field operation.
- Performed sensor calibration by cross-validating temperature and wind speed readings against standard measurement instruments, ensuring data accuracy.
- Collaborated directly with a Geomatics Engineering lecturer and the Geodesy Laboratory, bridging electrical/IoT engineering with geospatial science application.

Outcome

Delivered a functional, low-cost weather monitoring prototype with calibrated temperature and wind speed readings, validated through direct comparison testing against reference measurement instruments — demonstrating reliable performance for field deployment at a fraction of commercial system cost.

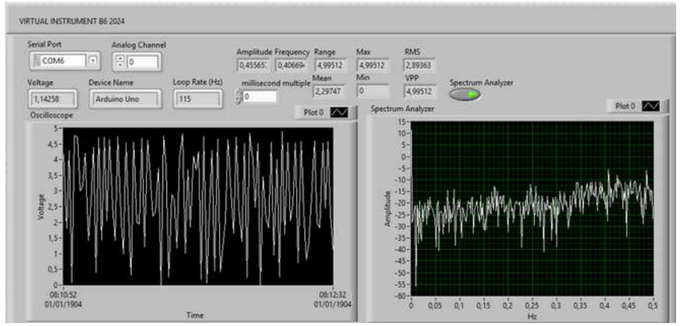
Design & Realization of an Integrated Multi-Module Electronics Trainer Kit Based on Virtual Instrumentation & System Enclosure

Kiicad | Proteus | Multisim | LabVIEW | Arduino Uno | Arduino IDE | CorelDraw

Motor Driver & Power Switching | Enclosure & Prototyping Design | Analog & Power Electronics Design

Testing & Calibration | Virtual Instrumentation & DAQ

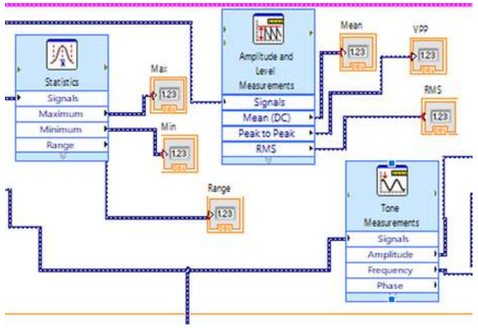




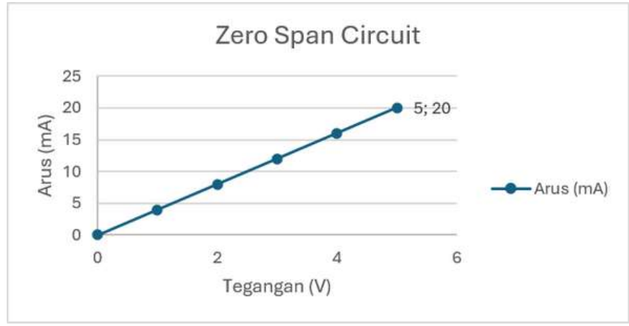














Description

An integrated electronics trainer kit combining 5 core sub-systems - analog electronics, power electronics, and data acquisition - into a single structured enclosure box, designed end-to-end: an adjustable symmetric power supply, an Op-Amp-based signal conditioning circuit, transistor bias analysis, a DC motor H-Bridge driver, and a real-time Virtual Instrument interface.

Key Insight

- Integrated 5 complex electronics modules into one ergonomic system enclosure, complete with fuse protection, heatsink cooling, and a digital voltmeter display.
- Designed a high-efficiency DC motor H-Bridge driver combining NE555 duty-cycle generation, TC4427CPA gate driver signal amplification, and power MOSFETs IRF3205 (N-Ch) / IRF5210 (P-Ch).
- Implemented an analog signal conditioning circuit (Op-Amp LM358N - voltage divider, voltage follower, current control) with measured accuracy testing, achieving very low relative error.
- Built a real-time Virtual Instrument in LabVIEW, communicating serially with an Arduino Uno R3 to display live spectrum analysis, duty cycle, and waveform generation.

Outcome

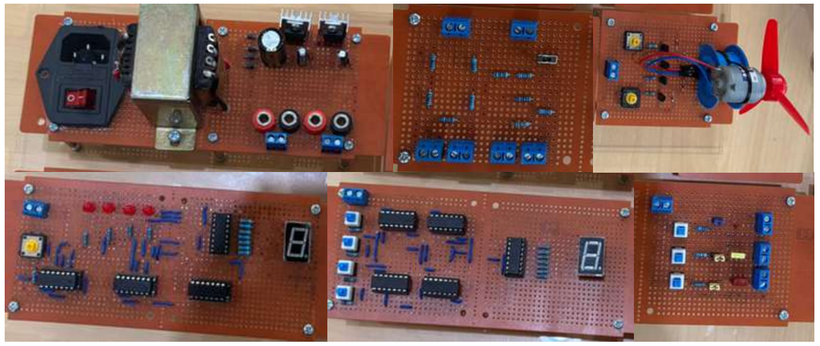
Successfully restructured and realized 1 fully functional and safe integrated Electronics Trainer Kit — proven to deliver high signal measurement accuracy and stable output voltage, ready for use as an interactive lab module in analog electronics, motor control, and computerized instrumentation.

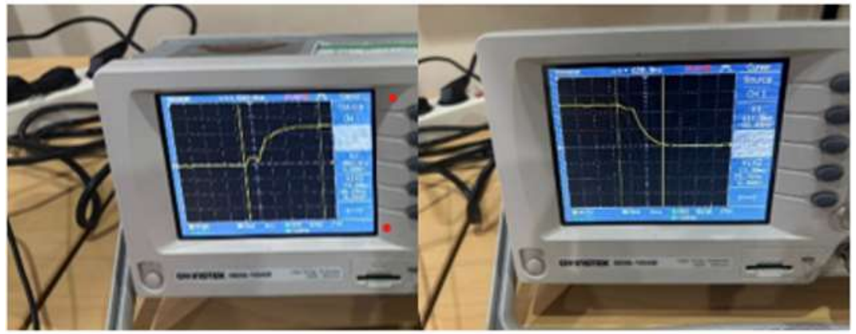
Design & Realization of an Integrated Basic Electronics & Digital Logic Trainer Kit Based on Case-Based Learning

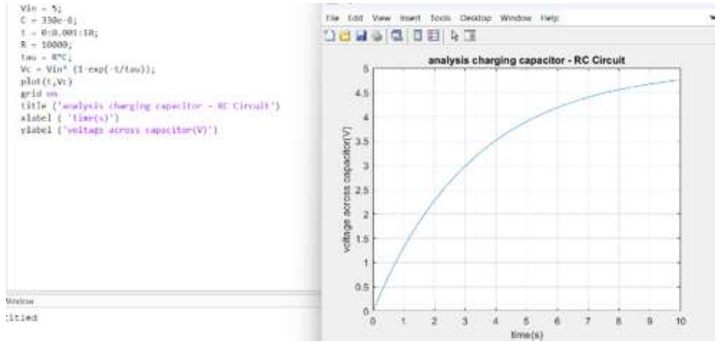
Multimeter | KiCad | Proteus | NI Multisim | Digital Oscilloscope

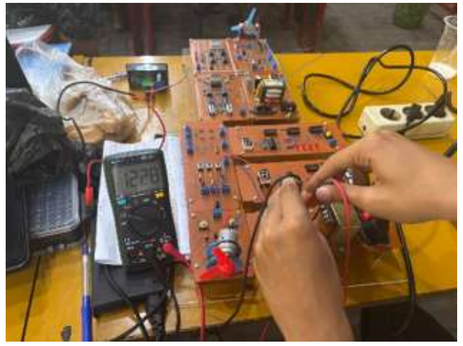
Digital Logic Design & Optimization | PCB Etching & Soldering | Analog Circuit Analysis

Signal/Waveform Measurement | Team Work | Sequential Logic Circuits | Troubleshooting

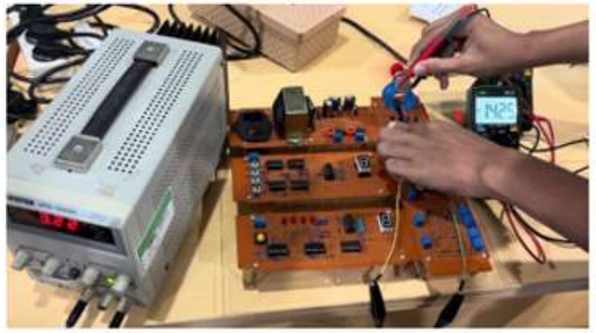




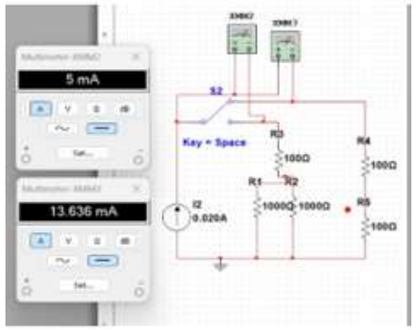












Description

A Case-Based Learning trainer module design and fabrication project for the Electrical Automation Engineering Department, integrating 6 fundamental circuit modules into one experimental board: a dual-regulated non-symmetric power supply (+12V/+5V), voltage/current divider circuits, an RC transient-response circuit, a 2-bit binary adder built from universal NAND gates, a 1-digit counter using JK flip-flops, and a DC motor direction control circuit.

Key Insight

- Designed a regulated non-symmetric power supply (full-wave rectifier + capacitive filter) delivering ripple-free +12V and +5V outputs for sensitive digital components, using LM7812/LM7805 regulators with fuse protection.
- Measured and verified RC charging/discharging waveforms across three component value variations (1kΩ/0.47μF, 680Ω/0.1μF, 10kΩ/0.33μF) on an oscilloscope, confirming the time constant ($\tau = R \times C$) against theoretical values.
- Optimized a 2-bit adder circuit by converting basic logic functions into a universal NAND gate (IC 7400) configuration, reducing chip count without sacrificing logic accuracy.
- Designed a DC motor direction control system (CW/CCW) using SPDT/DPDT switches and BC547/BC557 switching transistors, measuring the effect of input voltage variation (3V vs 6V) on motor speed and load current.

Outcome

Successfully designed, fabricated, and realized a fully functional Basic Electronics & Digital Logic Trainer Kit meeting 100% of technical specifications — all sub-modules passed practical testing with a very low relative error between theoretical calculation, software simulation (Proteus/Multisim), and real laboratory measurement.

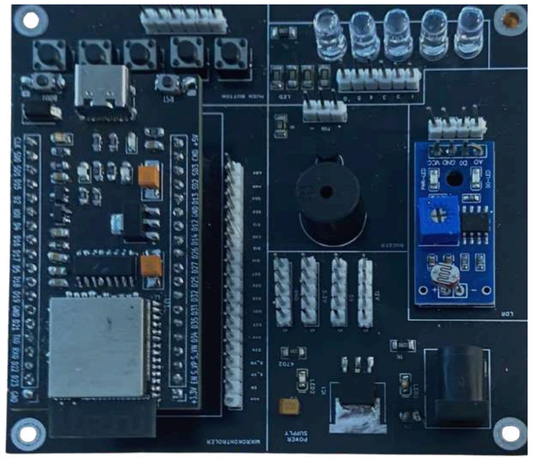
End-to-End Development of a Custom IoT Trainer Kit & Multi-Tiered Training Program (Basic, Intermediate, Advanced + AI Integration)

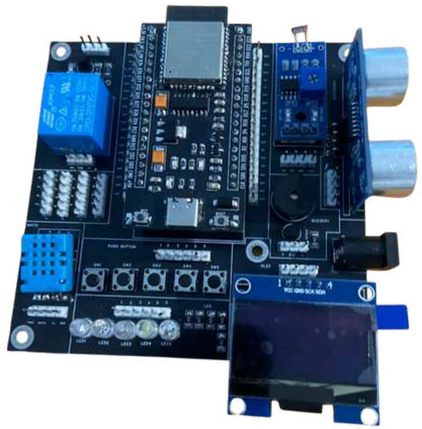
Fusion 360 (laser-cut acrylic enclosure) | ESP32 | DHT11 | LDR | MPU-6050 IMU | KiCad/EasyEDA

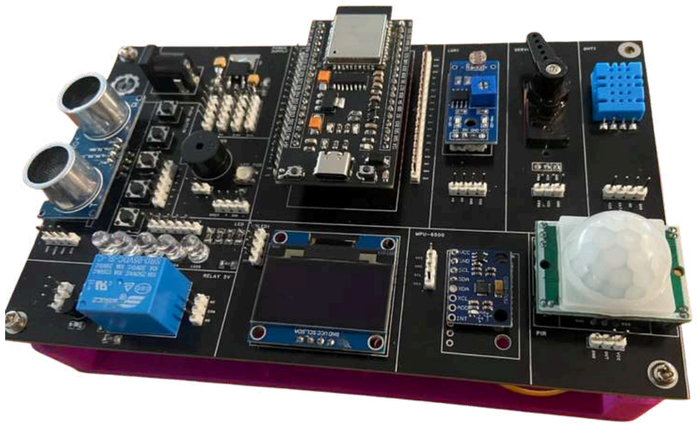
CorelDraw | HC-SR04 | Custom 2-Layer PCB | OLED SH1106 | Google Gemini API | Blynk IoT Cloud

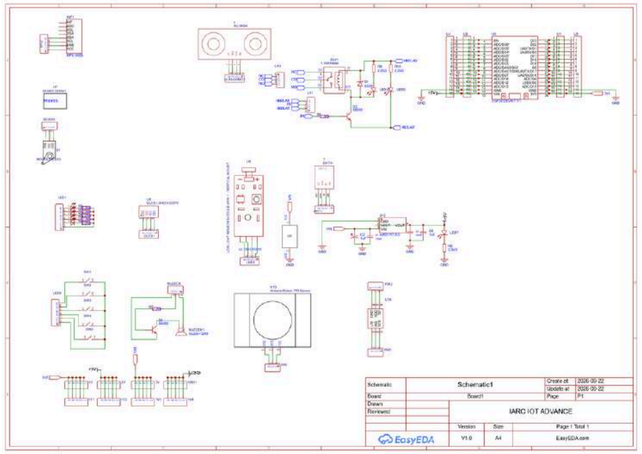
Multi-Sensor & Display Integration | Embedded Firmware | AI/LLM Integration (AIoT) | IoT Cloud Communication

Curriculum Design & Technical Training Delivery | Custom Hardware Design & Prototyping









Description

A comprehensive IoT training ecosystem combining the design and manufacturing of a custom IoT Trainer Kit (custom PCB and laser-cut acrylic enclosure) with a multi-tiered training program, structured from embedded systems fundamentals to AI integration:

- Basic Tier (4 hours): ESP32 fundamentals (Wi-Fi/Bluetooth, GPIO, ADC), LDR sensor reading via voltage divider, basic actuator control (LED and buzzer), and a simple monitoring app on Blynk IoT Cloud.
- Intermediate Tier (4 hours): Multi-sensor reading (HC-SR04, Raindrop, DHT11, LDR), local dashboard display on OLED/LCD I2C, relay and button control, non-blocking programming with millis(), and data sync to Blynk Cloud via Virtual Pins.
- Advanced Tier (4 hours): MPU-6050 IMU motion sensing, IoT-AI architecture, Google Gemini API integration on ESP32, and a Voice/Text Assistant for automated environmental reporting.

Key Insight

- Designed and manufactured in-house hardware, including a custom 2-layer PCB and laser-cut acrylic casing, for a plug-and-play, portable trainer kit.
- Structured a multi-tier curriculum from basic electronics to AI, enabling learners from diverse backgrounds to progress systematically.
- Implemented non-blocking firmware using millis() and BlynkTimer, keeping multi-sensor reading, display updates, and Wi-Fi communication stable and lag-free.
- Pioneered AIoT integration by connecting the Google Gemini API to an ESP32, enabling natural-language analysis of sensor data.

Outcome

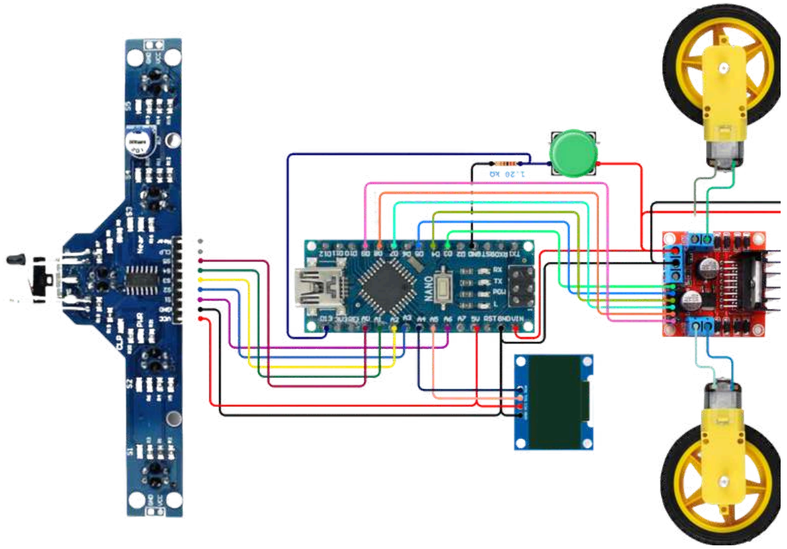
- Manufactured a ready-to-use IoT Trainer Kit for experimentation and prototyping.
- Delivered the full training program to 20+ participants per session, with a 100% completion rate.
- Produced a lab manual and a working Smart Environmental Monitoring and Voice Assistant prototype.

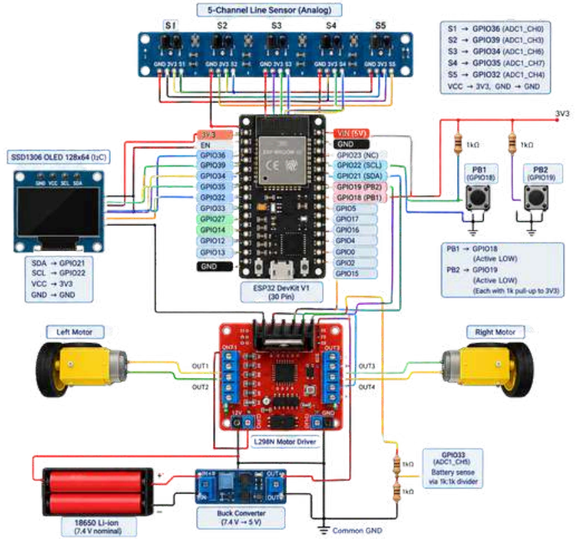
End-to-End Development of an Autonomous Line Follower Robot (LFM) Kit, Custom Web Simulator & Multi-Tiered Training Program

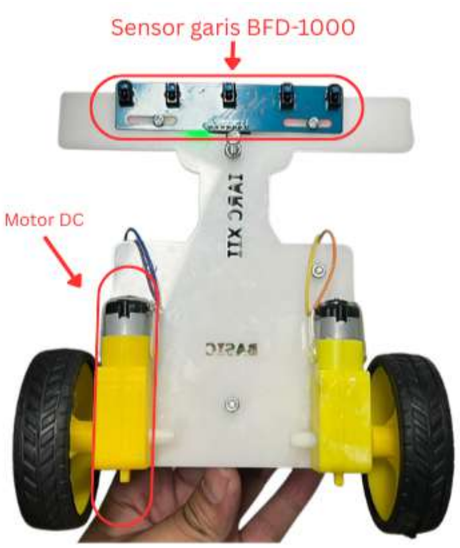
Arduino Nano | ESP32 | BFD-1000 | L298N Dual H-Bridge | OLED SSD1306 | PID Control

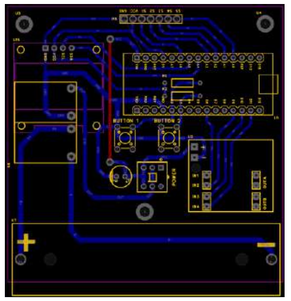
Autonomous Robotics & Embedded System Design | Hardware-Software Calibration & Debugging

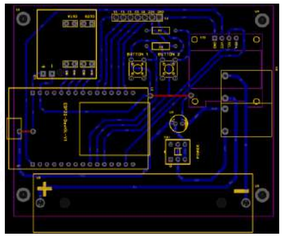
Web & Educational Software Engineering | Curriculum Design & Multi-Tier Pedagogy | PID Control & Tuning











Description

An integrated robotics and control-systems training program combining custom Line Follower Robot (LFM) hardware with original educational software, delivered in two tiers:

- Basic Tier (C/C++, Arduino Nano): Covers core components a 5-point IR sensor array, DC motor, and OLED display and basic navigation logic, including line tracking, turning, and lost-line recovery, through manual calibration.
- Advanced Tier (ESP32 and LF Blockly): Adds Wi-Fi auto-calibration, checkpoint navigation, and PID control, programmed through LF Blockly, a custom web application with a 2D simulator, live telemetry, PID tuning, and OTA updates.

Key Insight

- Built LF Blockly, an offline web application that lets users test code on screen before sending it to the robot over Wi-Fi.
- Connected basic C/C++ programming with visual block programming and PID tuning to guide users toward competition-level speed.
- Added live telemetry and real-time PID tuning (Kp, Ki, Kd), reducing test time from minutes to seconds.
- Built self-checking hardware with startup tests, lost-line protection, auto-calibration, and an OLED status display.

Outcome

- Delivered a stable LFM robot using Arduino Nano and ESP32.
- Released the complete LF Blockly platform, including a simulator, track editor, and OTA update system.
- Ran a training program that improved participants’ understanding of control systems and programming.

Thank You